INCH-POUND

MIL-PRF-1/1099H
16 June 2004
SUPERSEDING
MIL-PRF-1/1099G
18 July 1997

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON
TYPES 5948A, 5948B, AND 5948C

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Triode, hydrogen
See figures 1, 2, 3, and 4
Mounting position: Vertical, base down
Weight: 4.6 pounds (2.1 kg) nominal

ABSOLUTE RATINGS:

Parameter:	Ef V ac	epy kv	epx kv	Ebb Kv dc	Ip A ac	egy v	egx v	Ecc V dc	Ib a
Maximum:	6.6	25.0 1/	25.0 2/	---	30	---	650	---	1,000
Minimum:	6.0	10.0	5% epy	5.0	---	3/	---	0	---
Test conditions:	6.3	25.0	---	---	---	550	---	0	---

Parameter:	Ib A dc	prr	tj µs	pb	tk sec	TA °C	dik/dt a/µs	Eres V ac	Ires A ac
Maximum:	1.0	---	0.005 4/	9.0×10^9	---	+75	5,000	5.5 5/	8.0
Minimum:	---	---	---	---	600	-55	---	2.5	---
Test conditions:	---	360	---	---	600	---	---	6/	---

1/ Instantaneous starting is not recommended. When it is absolutely necessary, however, the maximum permissible epy is 18.0 kv and shall not be attained in less than 0.04 seconds.
2/ The peak inverse voltage, exclusive of a spike of 0.05 µs maximum duration, shall not exceed 5 kv during the first 25 µs following the anode pulse.
3/ The driver pulse, measured at the tube socket with thyratron grid disconnected, shall be: egy = 700 volts minimum, 1,000 volts maximum; tr = 0.35 µs maximum; tp = 2.0 µs maximum; impedance of driver circuit 50 to 200 ohms.
4/ Rated time jitter of 0.005 µs applies at a peak forward voltage of 15.0 kv or greater. At peak forward anode voltages between 10 and 15 kv, the tube is rated at a time jitter of 0.01 µs.
5/ The optimum reservoir voltage for operation in accordance with operation (1) conditions is inscribed on the base of the tube and must be held to within ± 5 percent.
6/ Adjust reservoir voltage to value indicated on tube to within ± 5 percent.

GENERAL: Qualification - Required

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TABLE I. Group A inspection.

Inspection	MIL-STD-1311 Method	Conditions	Acceptance level 9/	Symbol	Limits min	Limits max	Limits Unit
Conformance inspection, part 1							
Heater current	3241		0.65	If	14.0	33.0	Aac
Heater current	3241	Eres = 4.5 Vac	0.65	Ires	3.0	6.0	Aac
DC anode voltage for conduction	3241	Eres = inscribed value -5 percent	0.65	Ebb	----	4000	Vdc
Instantaneous starting 1/ 4/	3267	epy = 180 kv(min); Eres = inscribed value +5 percent	0.65	----	----	----	
Operation (1) 1/	3246	epy = 27.5 kv; Eres = inscribed value -5 percent; t = 30 minutes	0.65	egy	----	550	v
Operation (2) 5/ 6/	3246	epy = 15 kv; prr = 1,500; Eres = 95 percent of optimum value	0.65	egy	----	550	v
Operation of optimum reservoir voltage (1)	3246	Operation (1)	0.65	Eres	2.5	5.5	Vac
Operation of optimum 6/ reservoir voltage (2)	3246	Operation (2)	0.65	Eres	2.5	5.5	Vac
Pulse emission	3251	ik = 1,500 a; tp = 5.0 µs ±10 percent; prr = 60 ±10 percent; tr = 0.5 µs (max); Eres = inscribed value; starting time of current pulse = 2.5 µs	0.65	egk	----	400	v
Conformance inspection, part 2							
Operation (3)	3246	Operation (1), except Eres = inscribed value +5 percent; tk = 900 sec; t = 1,200 sec	----	egy	----	550	v
Operation (4)	3246	Operation (2), except Eres = 105 percent; optimum value	----	egy	----	550	v
Anode delay time	3256	Operation (1); t = 120 sec	----	tad	----	1.0	µs
Anode delay time drift 7/	3256	Anode delay time	----	–tad	----	0.25	µs
Time jitter 8/	3261	Operation (1); except epy = 12.5 kv	----	tj	----	0.004	µs
Conformance inspection, part 3							
Life test (1) 1/	----	Group C; t = 96 hours "on" and 1 hour "off"; t = 500 hours	----	----	----	----	----

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TABLE I. Group A inspection.

Inspection	MIL-STD-1311 Method	Conditions	Acceptance level 9/	Limits Symbol	Limits min	Limits max	Unit
Conformance inspection, part 3 -Continued							
Life test (2) 6/	----	Life test (1); operation (2) Zg = 250 Ω; t = 500 hours	----	----	----	----	----
Life test (1) and (2) endpoints:							
Operation (1)	3246		----	egy	----	650	v
DC anode voltage for conduction	3247	egy = 500 v	----	Ebb	----	4500	Vdc
Time Jitter	3261	egy = 500 v; Zg = 250 Ω	----	tj	----	0.005	µs
Operation at elevated ambient temperature 1/ 2/	3246	TA = 75°C; t = 5 hours	----	egy	----	550	v
Life test (3) 1/ 3/	----	t = 500 hours	----	----	----	----	----
Life test (3) endpoints:							
Operation (1)	3246		----	egy	----	675	v
DC anode voltage for conduction	3247		----	Ebb	----	4500	Vdc
Time Jitter	3261	egy = 650 v	----	tj	----	0.01	µs

1/ The circuit constants shall be chosen so that at epy = 25.0 kv under charging conditions: dik/dt = 5,000 a/µs minimum; ib = 1,000 a minimum; tp = 2.5 ± 0.25µs; prr = 500 minimum. The grid pulse characteristics shall be: tp = 2.0 µ maximum; tr = 0.35 µs minimum; internal impedance of driver = 250 ohms minimum.

2/ This test shall be conducted for a total of 5 consecutive hours with no more than three kickouts and with no evidence of anode heating. The tube shall be started at Eres = 105 percent of the inscribed value and operate at this value for 4 hours. At the start of the fifth hour and while the tube is still operating, the reservoir voltage shall be lowered to 95 percent of the inscribed value and remain there for the final 1 hour of operation.

3/ The life test shall be intermittent, with the tube operating approximately half the total elapsed time, and held without reservoir voltage or cathode-heater voltage approximately half the total time. The tube shall make a minimum of four starts during each 24-hour period. The life-test duration is measured in operating hours. Failure of the tube to operate following the minimum allowable cathode and reservoir heater warmup time constitutes failure of the test.

4/ The tube shall operate satisfactorily on push-button starting within three attempts when the anode voltage (epy) is applied to the tube under test in such a manner as to rise from 0 to 18 kv within 0.03 second. (The filter in the rectifier shall be designed so that the epy reaches at least 9 kv within 0.015 second.) Any tube failing to start within three attempts will be considered a failure.

5/ The anode circuit constants shall be so chosen that a resonant charging conditions; epy = 15 kv; ib = 500 a minimum; dik/dt = 5,000 a/µs minimum. Output pulse; tp = 1.3 µs ±10 percent; prr = 1,500 minimum; epx = 5.0 kv. Grid pulse the same as note 1. Adjust reservoir voltage to optimum value for these test conditions. The tube shall operate continuously for 20 minutes without continuous conduction.

6/ The optimum reservoir voltage for operation in accordance with operation (2) conditions shall be so determined that the tube shall operate at 95 to 105 percent of the established reservoir voltage for these conditions.

7/ This test shall be performed simultaneously with the operation (1) test. An anode delay time measurement shall be made at the end of 2 minutes, 4 minutes, and 30 minutes of the operation (1) test. The –tad measurement is the numerical difference between the 2-minute and 4-minute, or 2-minute and 30-minute anode delay time readings, whichever is greater.

8/ The tj shall be not greater than the amount specified after 120 seconds of anode operation.

9/ This specification sheet uses accept on zero defect sampling in accordance with MIL-PRF-1, table III.

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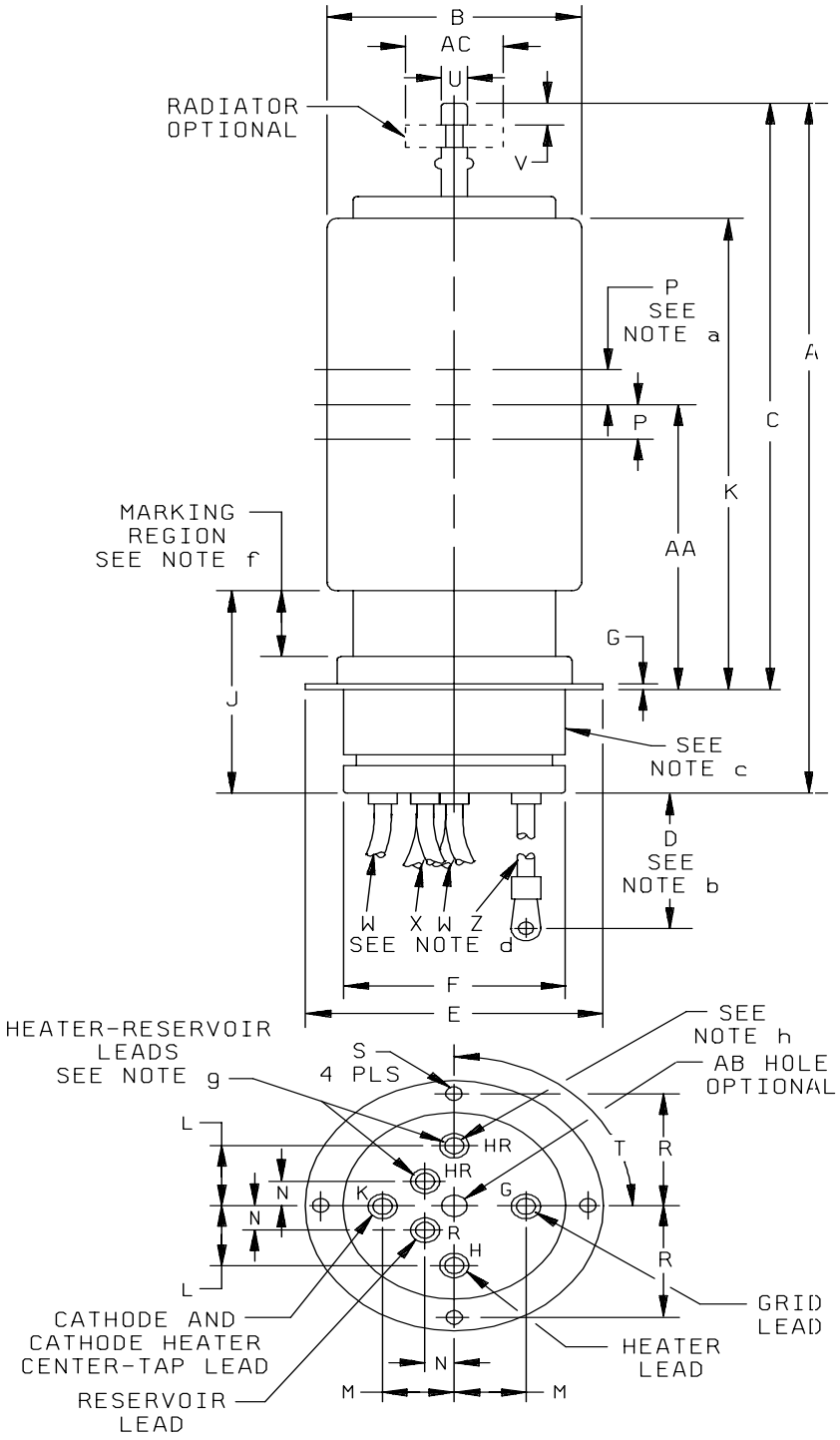


FIGURE 1. Outline drawing of electron tube type 5948A (glass version).

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LTR	Dimensions			
	Inches		Millimeters	
Conformance inspection, part 2				
	Min	Max	Min	Max
A	15.250	16.250	387.35	412.75
B	4.875	5.125	123.83	130.18
C	13.125	13.875	333.38	352.42
F	4.438	4.562	112.73	115.87
R	2.662	2.682	67.61	68.12
T	89.6°		90.4°	
Conformance inspection, part 3				
See note k				
D	5.000	6.000	127.00	152.40
E	5.938	6.062	150.83	153.97
J	4.344	4.656	110.34	118.26
K	9.938	11.062	252.43	280.97
L	1.438	1.562	36.53	39.67
M	1.312	1.438	33.32	36.53
N	0.438	0.562	11.13	14.27
S	0.309	0.315	7.85	8.00
U	0.559	0.573	14.20	14.55
V	0.500		12.70	
AC	-----	2.000	-----	50.80
P	-----	1.000	-----	25.40
AA	-----	6.500	-----	175.10
Reference dimensions				
G	0.094		2.39	
AB	0.500		12.70	
W	4,500 (circular mils)			
X	1,500 (circular mils)			
Z	2,000 (circular mils)			

NOTES:

a. Dimension P designates the area to be used for clamping.
b. Dimension D is measured from bottom of base insert to center of lead lug hole.
c. Base, metal shell, with mounting flange and phenolic insert. Flange secured to base with rivets, or equivalent. Insert shall be low-loss phenolic wafer or laminated phenolic resin, permanently marked as follows: G (grid), H (heater), R (reservoir), HR (heater-reservoir), and K (cathode). All feed-through fittings shall be equipped with setscrews, or equivalent.
d. Leads shall be flexible. Dimensions W, X, and Z apply to cross-sectional area. Lead identification insulation sleeving and lugs shall be as follows:

Lead	Color	Sleeving	Lugs (see note j)
G (grid)	Green	H-A-1(see note e)	SAE-AS25036-108
H (heater)	Yellow	H-A-1 or H-B-1	SAE-A25036-157
HR (heater) (see note g)	Yellow with black sleeve	H-A-1 or H-B-1	SAE-AS25036-157
HR (reservoir) (see note g)	Red with yellow sleeve	H-A-1 or H-B-1	SAE-AS25036-108
R (reservoir)	Red	H-A-1 or H-B-1	SAE-AS25036-108
K (cathode)	Black	H-A-1 or H-B-1	SAE-AS25036-157

e. As specified in specification MIL-I-3190.
f. Reservoir voltage shall be inscribed once above each mounting hole on base wall.
g. Heater lead internally connected to reservoir lead.
h. Cathode heater connection to this lead.
j. Lugs shall be permanently identified with symbols as follows: G (grid), K (cathode), H (heater), R (reservoir), and HR (heater-reservoir).
k. Holes S shall be inspected for orientation by using the base gage outlined in figure 4.

FIGURE 1. Outline drawing of electron tube type 5948A (glass version) - Continued.

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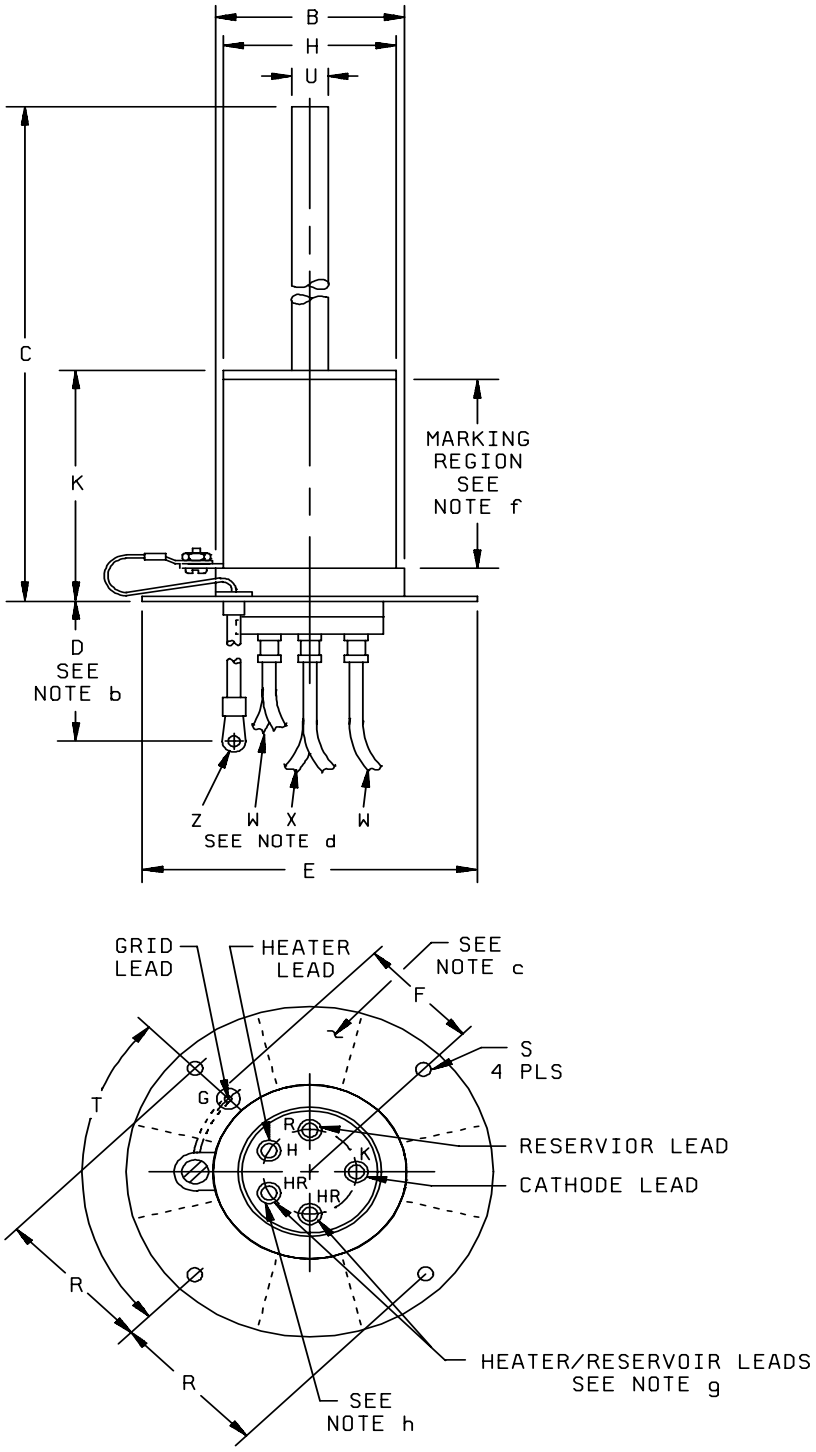


FIGURE 2. Outline drawing of electron tube type 5948B (ceramic verison).

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LTR	Dimensions			
	Inches		Millimeters	
Conformance inspection, part 2				
	Min	Max	Min	Max
B	3.125	3.375	79.37	85.73
C	12.532	13.875	318.31	352.43
F	------	2.281	-----	57.94
R	2.662	2.682	67.61	68.12
T	89.6°	90.4°	----	----
Conformance inspection, part 3				
See note k				
D	8.000	9.000	203.20	228.60
E	5.938	6.062	150.83	153.97
H	2.875	3.125	73.03	79.38
K	3.925	5.000	99.70	127.00
S	0.309	0.315	7.85	8.00
U	0.552	0.572	14.02	14.53
Reference dimensions				
W	4,500 (circular mils)			
X	1,500 (circular mils)			
Z	2,000 (circular mils)			

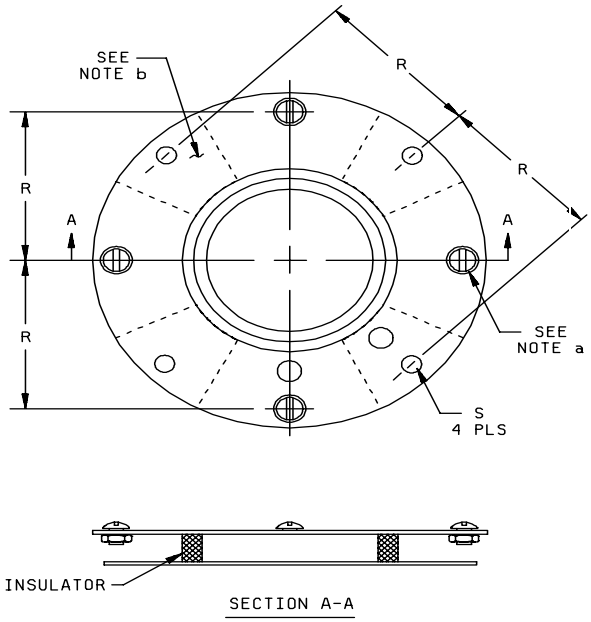
NOTES:
a. A socket adapter (figure 3) is supplied with each thyratron for applications that required electrical isolation of the thyratron mounting flange from the cathode (K) lead.
b. Dimension D is measured from bottom of flange to center of lead lug hole.
c. Flange cutouts are optional.
d. Leads shall be flexible. Leads shall be identified as follows:

Lead	Color	Sleeving	Lugs (see note j)
G (grid)	Green	H-A-1(see note e)	SAE-AS25036-108
H (heater)	Yellow	H-A-1 or H-B-1	SAE-AS25036-157
HR (heater) (see note g)	Yellow with black sleeve	H-A-1 or H-B-1	SAE-AS25036-157
HR (reservoir) (see note g)	Red with yellow sleeve	H-A-1 or H-B-1	SAE-AS25036-108
R (reservoir)	Red	H-A-1 or H-B-1	SAE-AS25036-108
K (cathode)	Black	H-A-1 or H-B-1	SAE-AS25036-157

e. As specified in specification MIL-I-3190.
f. Reservoir voltage shall be inscribed once above each mounting hole on base wall.
g. Heater lead internally connected to reservoir lead.
h. Cathode heater connection to this lead.
j. Lugs shall be permanently identified with symbols as follows:
G (grid), K (cathode), H (heater), R (reservoir), and HR (heater-reservoir).
k. Holes S shall be inspected for orientation by using the base gage outlined in figure 4.

FIGURE 2. Outline drawing of electron tube type 5948B (ceramic version) - Continued.

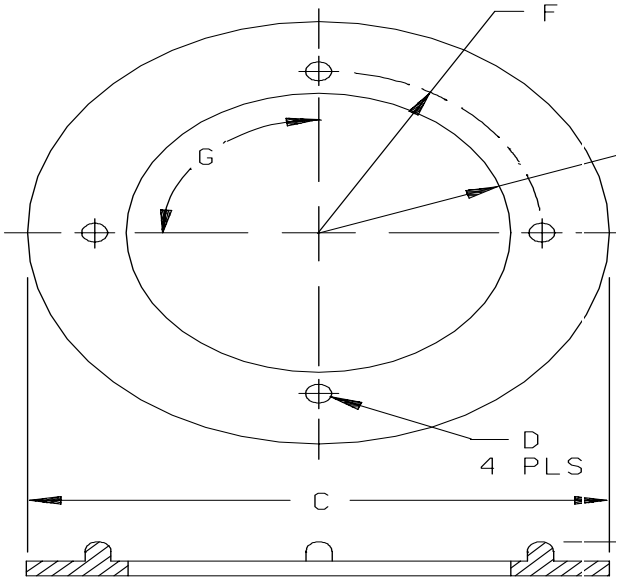
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Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
R	2.662	2.682	67.61	68.12
S	.309	.315	7.85	8.00

NOTE:
a. 10-32 X .250 truss head machine screws.
b. Flange cutouts are optional.

FIGURE 3. Socket adapter.



Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.238	.262	6.05	6.65
B	.296	.328	7.52	8.33
C	6.650	7.350	168.91	186.69
D	0.2765 DIA	0.2775 DIA	7.02 DIA	7.05 DIA
E	2.312 R	2.314 R	58.72 R	58.76 R
F	2.671 R	2.673 R	67.84 R	67.89 R
G	89.08°	90.02°	-----	-----

FIGURE 4. Base gage.

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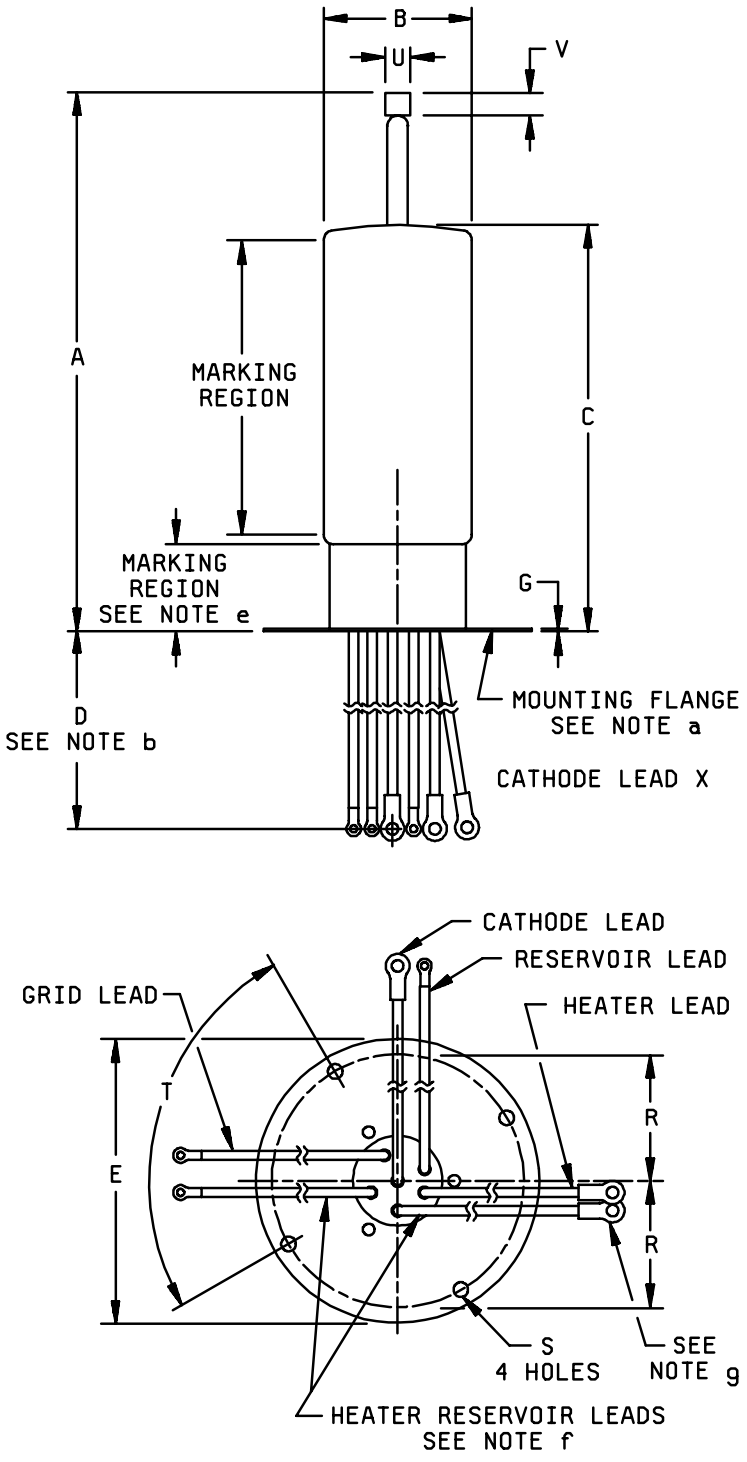


FIGURE 5. Outline drawing of electron tube type 5948C (alternate glass version).

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Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
Conformance inspection, part 2				
A	11.530	12.281	292.87	311.93
B	---	3.312	---	84.12
C	8.343	9.469	211.90	240.50
R	2.662	2.682	67.61	68.12
T	89°60'	90°40'	89°60'	90°40'
Conformance inspection, part 3 (see note j)				
D	7.500	8.500	190.50	215.90
E	5.938	6.062	150.83	153.97
S	.309	.315	7.85	8.00
U	.559	.573	14.20	14.55
V	.500	---	12.70	---
Reference dimensions				
G	.063		1.60	
W	14 AWG		2.5 mm^2	
X	10 AWG		6.0 mm^2	

NOTES:

a. Mounting flange is electrically isolated.
b. Dimension D is measured from bottom of flange to center of lead lug hole.
c. Leads shall be flexible. Leads shall be identified as follows:

Lead	Color	Sleeving	Reference dimension	Lug (see note h)
G (grid)	Green	H-A-1 (see note d)	W	SAE-AS25036-108
H (heater)	Yellow	H-A-1 or H-8-1	W	SAE-AS25036-157
HR (heater) (see note f)	Yellow with black sleeve	H-A-1 or H-8-1	W	SAE-AS25036-157
HR (reservoir) (see note f)	Red with yellow sleeve	H-A-1 or H-8-1	W	SAE-AS25036-108
R (reservoir)	Red	H-A-1 or H-8-1	W	SAE-AS25036-108
K (cathode)	Black	H-A-1 or H-8-1	X	SAE-AS25036-157

d. As specified in MIL-I-3190.
e. Reservoir voltage shall be inscribed once above each mounting hole on aluminum base shell.
f. Heater lead HR internally connected to reservoir lead HR.
g. Cathode heater connection to this lead.
h. Lugs shall be permanently identified with symbols as follows:
G (grid), K (cathode), H (heater), R (reservoir), and HR (heater-reservoir).
i. Holes shall be inspected for orientation by using the base gauge outlined in figure 4.

FIGURE 5. Outline drawing of electron tube type 5948C (alternate glass version) - Continued.

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NOTES

Referenced documents. In addition to MIL-PRF-1, this specification sheet references MIL-STD-1311, MIL-I-3190 and SAE-AS25036.

Changes from previous issue. The margins of this specification sheet are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force -11
DLA -CC

Preparing activity:
DLA - CC

(Project 5960-3718)

Review activities:
Navy - AS, CG, MC, OS
Air Force - 19, 99

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